EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

The **Impulse Invariance Method** converts an analog filter transfer function $H_a(s)$ into a digital filter $H(z)$ by directly sampling the continuous impulse response: $h[n] = T_d h_a(nT_d)$.

Pedagogical Strategy:

1. Derive the continuous-to-discrete pole mapping: An analog pole at $s = p_k$ maps directly to a discrete pole at $z = e^{p_k T_d}$.
2. Prove stability preservation: Left-half s-plane ($\mathrm{Re}(p_k) < 0$) maps strictly inside the unit circle ($|e^{p_k T_d}| < 1$).
3. Formulate the Partial Fraction Expansion transformation equation.
4. Highlight the **fundamental limitation of Impulse Invariance**: Spectral aliasing:

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_a\left(j\frac{\omega + 2\pi k}{T_d}\right)$$

5. Explain why Impulse Invariance is restricted *strictly* to bandlimited Lowpass and Bandpass filters, and **cannot** be used to design Highpass or Bandstop filters.

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the Impulse Invariance mapping formula for simple and repeated poles.
2. **Transform** continuous transfer functions $H_a(s)$ into discrete transfer functions $H(z)$.
3. **Analyze** the mechanism and severity of spectral aliasing.
4. **Determine** when Impulse Invariance is applicable in practical filter design.

2. MATHEMATICAL FOUNDATIONS

2.1 Pole Mapping Derivation

Given analog transfer function in partial fraction form:

$$H_a(s) = \sum_{k=1}^{N} \frac{A_k}{s - p_k}$$

The continuous impulse response is:

$$h_a(t) = \sum_{k=1}^{N} A_k e^{p_k t} u(t)$$

Sampling at $t = nT_d$ with gain scaling T_d:

$$h[n] = T_d h_a(nT_d) = T_d \sum_{k=1}^{N} A_k e^{p_k n T_d} u[n] = T_d \sum_{k=1}^{N} A_k (e^{p_k T_d})^n u[n]$$

Taking the Z-Transform of $h[n]$:

$$H(z) = T_d \sum_{k=1}^{N} \frac{A_k}{1 - e^{p_k T_d} z^{-1}}$$

2.2 Pole-Mapping Properties

- Let $p_k = \sigma_k + j\Omega_k$.
- Discrete pole: $z_k = e^{p_k T_d} = e^{\sigma_k T_d} e^{j\Omega_k T_d}$.
- **Magnitude:** $|z_k| = e^{\sigma_k T_d}$.
  - If $\sigma_k < 0$ (Left-half s-plane) $\implies$ $|z_k| < 1$ (Inside unit circle $\implies$ **Stable**).
  - If $\sigma_k = 0$ ($j\Omega$ axis) $\implies$ $|z_k| = 1$ (On unit circle).
  - If $\sigma_k > 0$ (Right-half s-plane) $\implies$ $|z_k| > 1$ (Outside unit circle $\implies$ **Unstable**).

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3. WORKED NUMERICAL EXAMPLES

Example 27.1: Impulse Invariance Transformation

Problem: Apply the Impulse Invariance method to design a digital filter from:

$$H_a(s) = \frac{s+1}{(s+1)^2 + 9} = \frac{s+1}{(s+1-3j)(s+1+3j)}$$

with sampling interval $T_d = 0.1$ s.

Solution:

1. **Partial Fraction Expansion:**

$$H_a(s) = \frac{0.5}{s+1-3j} + \frac{0.5}{s+1+3j}$$

   Poles: $p_1 = -1 + 3j, ; p_2 = -1 - 3j$.
2. **Apply Impulse Invariance Mapping:**

$$z_1 = e^{p_1 T_d} = e^{(-1+3j)(0.1)} = e^{-0.1} e^{j0.3} = 0.9048 e^{j0.3} = 0.9048(\cos 0.3 + j \sin 0.3) = 0.8644 + 0.2674j$$

$$H(z) = T_d \left[\frac{0.5}{1 - e^{p_1 T_d} z^{-1}} + \frac{0.5}{1 - e^{p_2 T_d} z^{-1}} \right] = 0.1 \left[\frac{1 - e^{-0.1} \cos(0.3) z^{-1}}{1 - 2e^{-0.1} \cos(0.3) z^{-1} + e^{-0.2} z^{-2}} \right]$$

$$H(z) = \frac{0.1 - 0.08644 z^{-1}}{1 - 1.7288 z^{-1} + 0.8187 z^{-2}}$$

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Derive the Impulse Invariance mapping equation. Explain why highpass filters cannot be designed using this method. *(8 Marks)* **(b)** Convert $H_a(s) = \frac{2}{(s+1)(s+2)}$ into a digital filter $H(z)$ using Impulse Invariance with $T_d = 1$ s. *(7 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
    - Mathematical derivation from $h_a(t) \to h[n] \to H(z)$ *(5 Marks)*
    - Explanation of periodic aliasing $\sum H_a(j(\omega + 2\pi k)/T_d)$ corrupting high frequencies *(3 Marks)*
- **Part (b):**
    - PFE: $H_a(s) = \frac{2}{s+1} - \frac{2}{s+2}$ *(2 Marks)*
    - Apply mapping with $T_d = 1$: $H(z) = 1 \cdot \left[\frac{2}{1-e^{-1}z^{-1}} - \frac{2}{1-e^{-2}z^{-1}}\right] = \frac{2}{1-0.3679z^{-1}} - \frac{2}{1-0.1353z^{-1}}$ *(3 Marks)*
    - Combine: $H(z) = \frac{2(1-0.1353z^{-1})-2(1-0.3679z^{-1})}{(1-0.3679z^{-1})(1-0.1353z^{-1})} = \frac{0.4652z^{-1}}{1-0.5032z^{-1}+0.0498z^{-2}}$ *(2 Marks)*

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5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

b = [2]
a = [1, 3, 2]
bz, az = signal.impinvar(b, a, fs=1.0)
print("Impulse Invariance Numerator:", np.round(bz, 4))
print("Impulse Invariance Denominator:", np.round(az, 4))
```